

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

spent all claude tokens
i need more coffee

no problem

good thanks and you

fine

alive no

nope

all good goona good

not yet

haven't started yet

Today's agenda

- Total Recall
- User Interface Design
- Discuss the task
- Figma quick start

Total Recall



What do you remember from the last session?

we can't hear you

i can hear you

same

java

figma

primary keys

database db

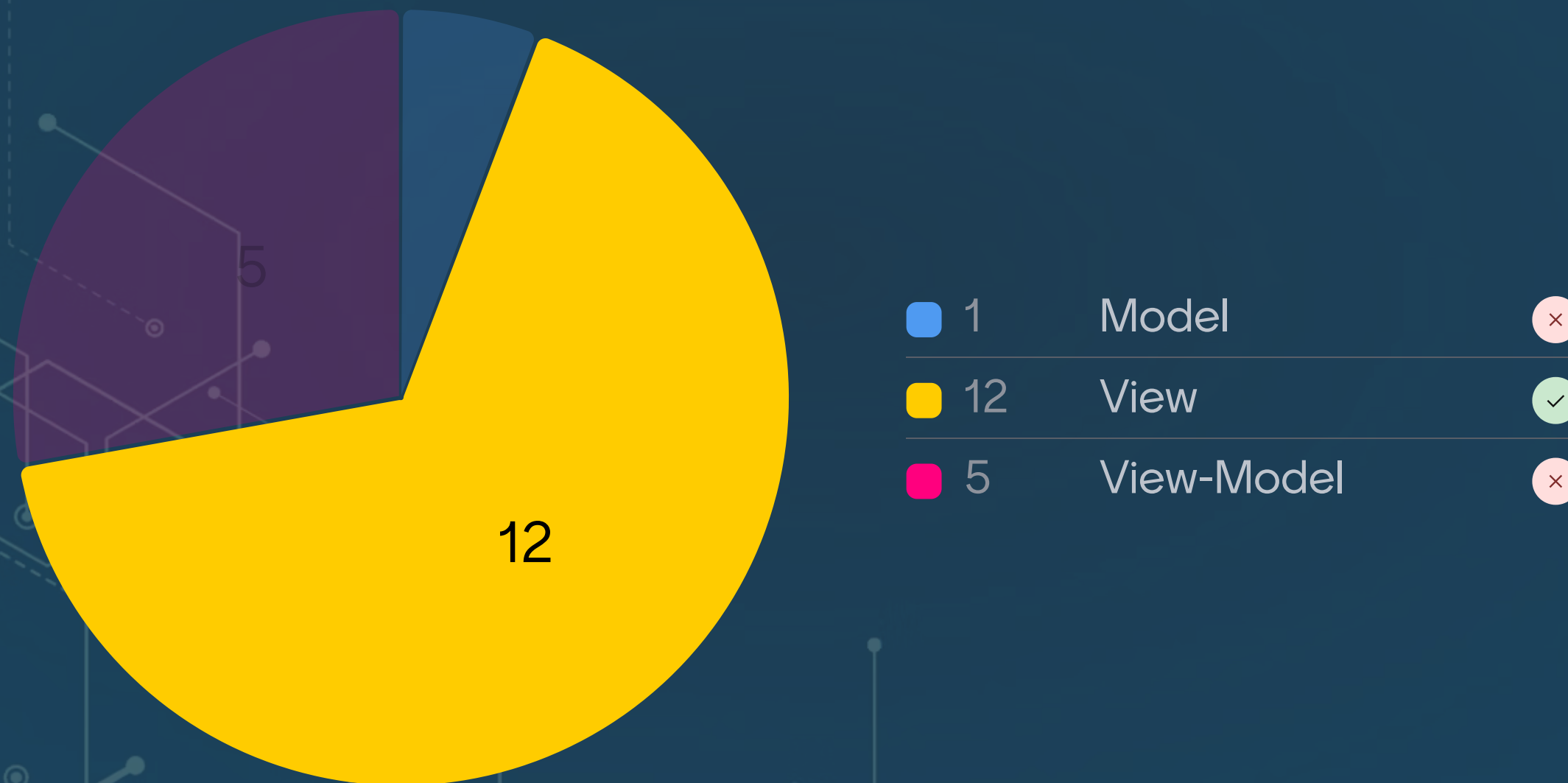
magic

database normalization

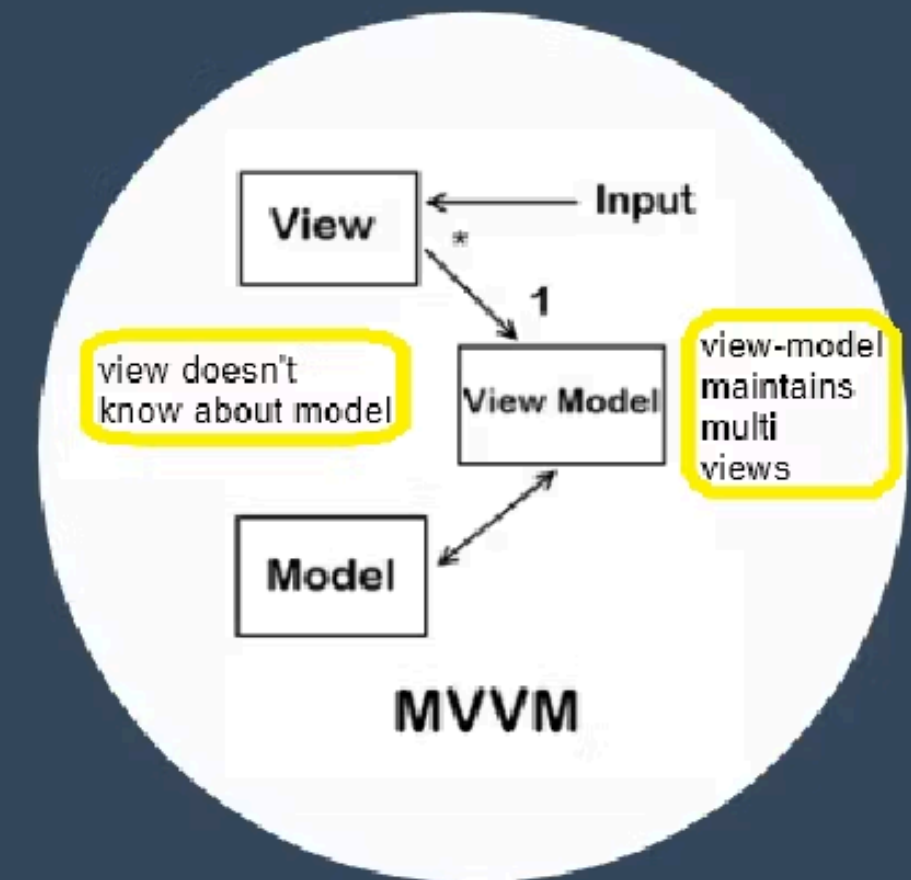
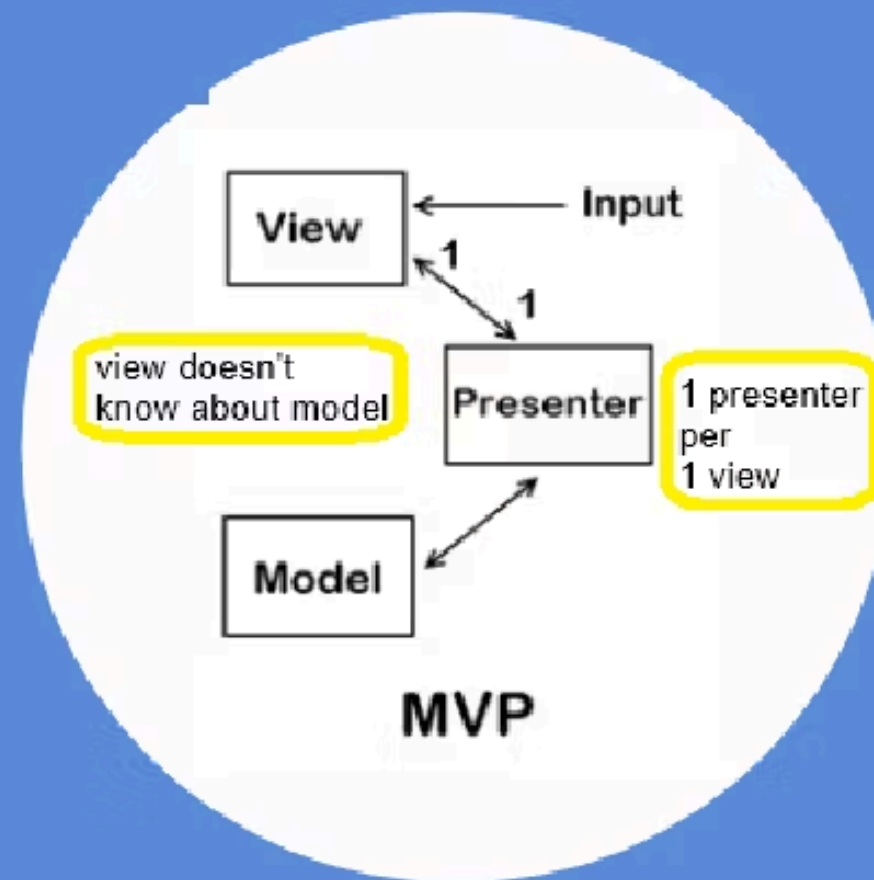
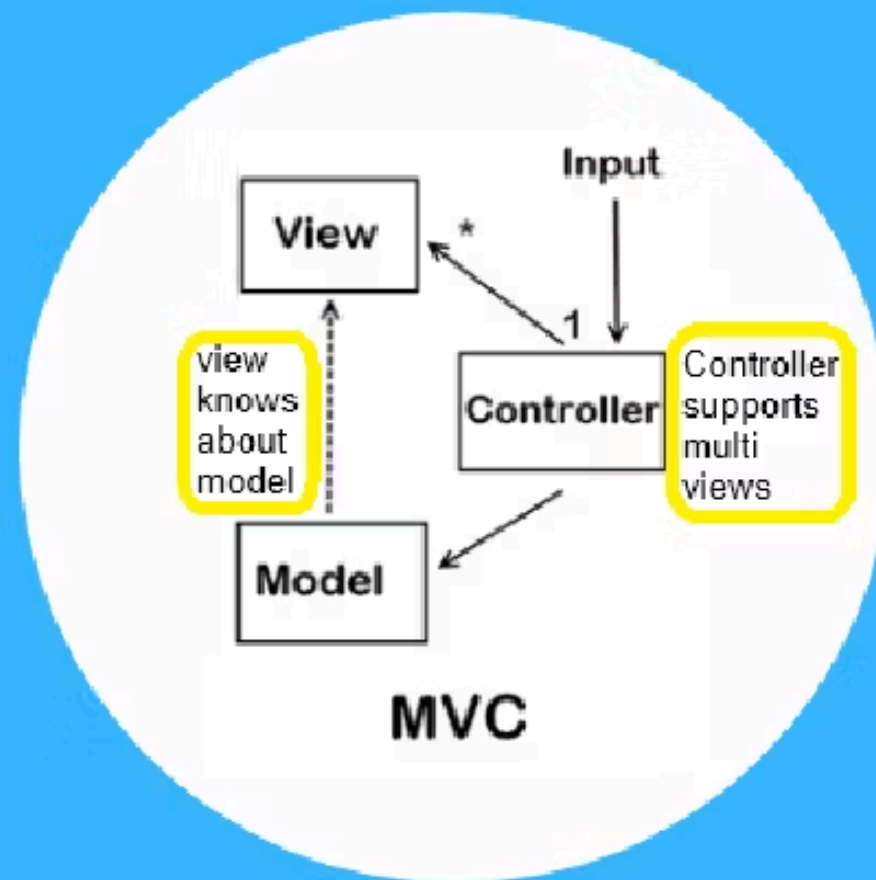
keys

primary key

Who is receiving requests in MVVM?



MVC VS MVP VS MVVM



ITPU IT PARK
UNIVERSITY

select correct meaning:



0 ✗

entity

0 ✗

weak entity

0 ✗

relationship

0 ✗

weak
relationship

19 ✓

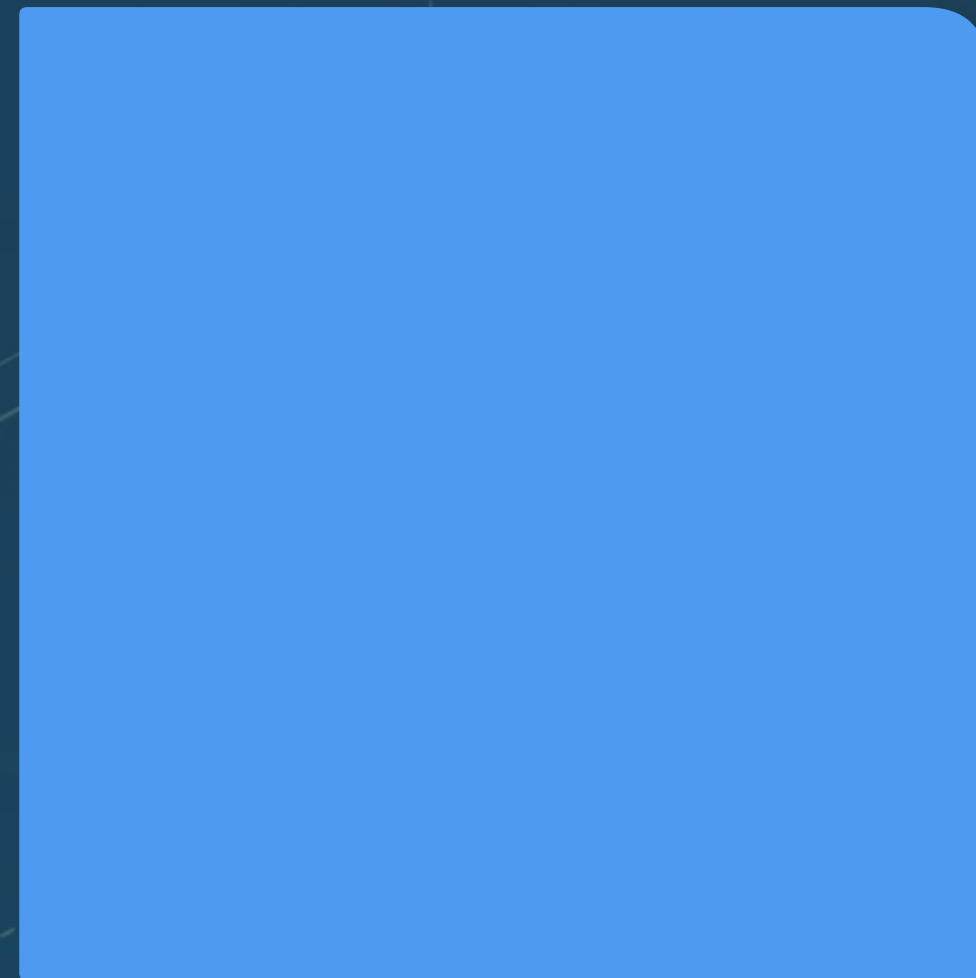
attribute

0 ✗

multivalued
attribute

What is used when two or more primary keys from another entities are necessary to determine a unique entity?

16 ✓



Compound key

4 ✗



Composite key

0 ✗

Relational key

select all true:

21 ✓



mandatory many

0 ✗

many to one

0 ✗

optional many



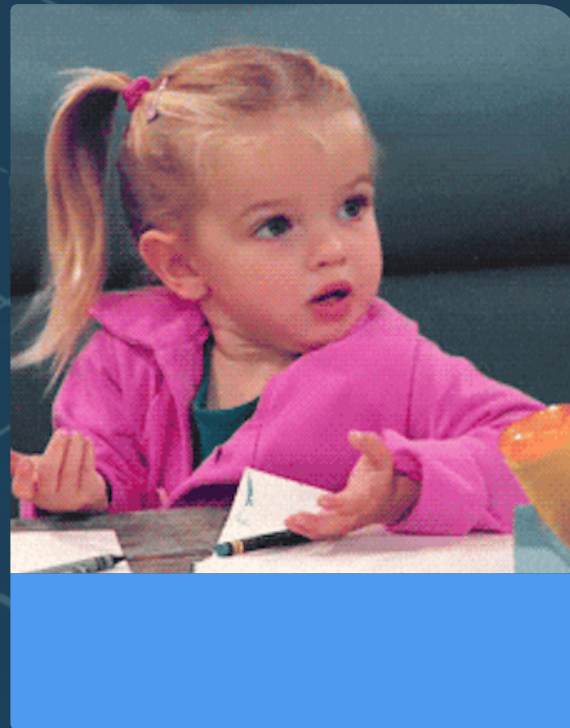
Architecture Advanced (ws-04)

UID (UI\UX)
User Interface Design

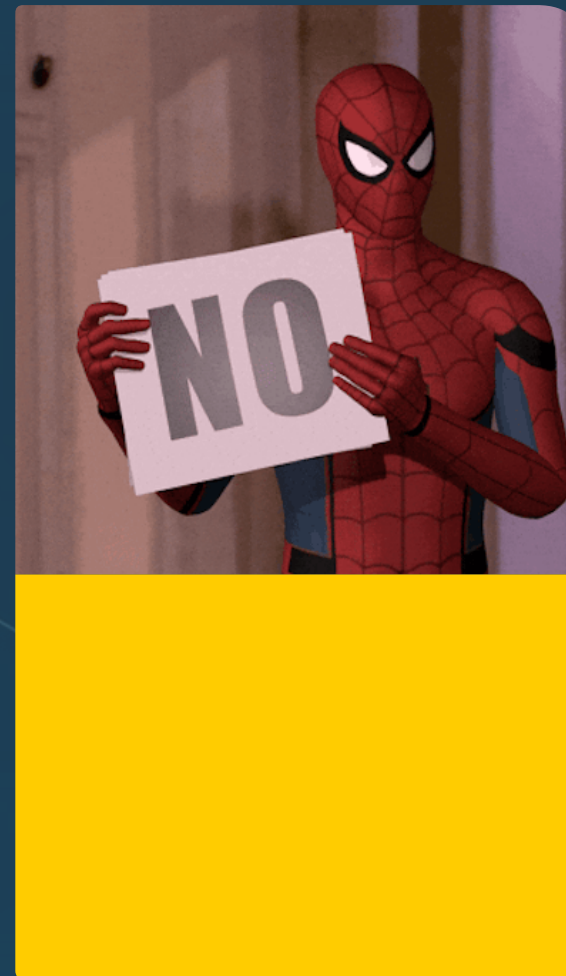
Have you started the theory of 'Web App Interface Design'?

11

4



????????????????



3



+

User Interface design

A well-known quote among UX designers states that "every dollar spent on UX brings in between *2and* 100 dollars in return". Why is that so?

Web apps with a confusing, incomprehensible UI fail to provide high retention and conversion rates. If your users don't find what they are looking for in the first few seconds, they will simply move to another website or app. That is why companies spend so much time and money on creating user-friendly interfaces. While working on your web app, you should think about common layouts, important interfaces, and how to navigate through them.

Evolution

As of 2019: <https://www.creativebloq.com/ux/3-way-create-website-mockups-11513936>

As of 2026:

- Figma: The industry standard for UI/UX design. Offers real-time team collaboration, robust design systems, and built-in AI layout generation.
- Framer: Best for turning high-fidelity visual mockups into fully responsive, published websites with native animations and micro-interactions.
- v0 by Vercel: An AI-native UI tool that generates clean React and Tailwind CSS code mockups directly from text prompts.
- UXPin: Ideal for advanced interactive mockups using production-ready code components and complex design logic.

trending

- **Replit** and **Lovable** are fundamentally **AI app builders / code environments**, not pure visual mockup tools like Figma.
- **Different primary use case:** They are built to generate full-stack, functional software, database backends, and live deployment. While tools like **v0** focus heavily on rapid visual UI/CSS mockups, Replit and Lovable lean into writing full application logic.
- **Workflow shift:** Designers use visual tools (Figma, Framer) to map out aesthetics, components, and layout flows before engineering. Using Replit or Lovable right off the bat means jumping straight to full software development.

*Note: Features like **Replit Design** or **Lovable's UI generator** blur this line, but their core strength remains end-to-end app creation rather than purely design-first UI prototyping.*

Explore ws-04

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026/tree/main/ws-04-architecture-advanced>

Quiz

What is a **key** principle in UI design to ensure a great user experience?

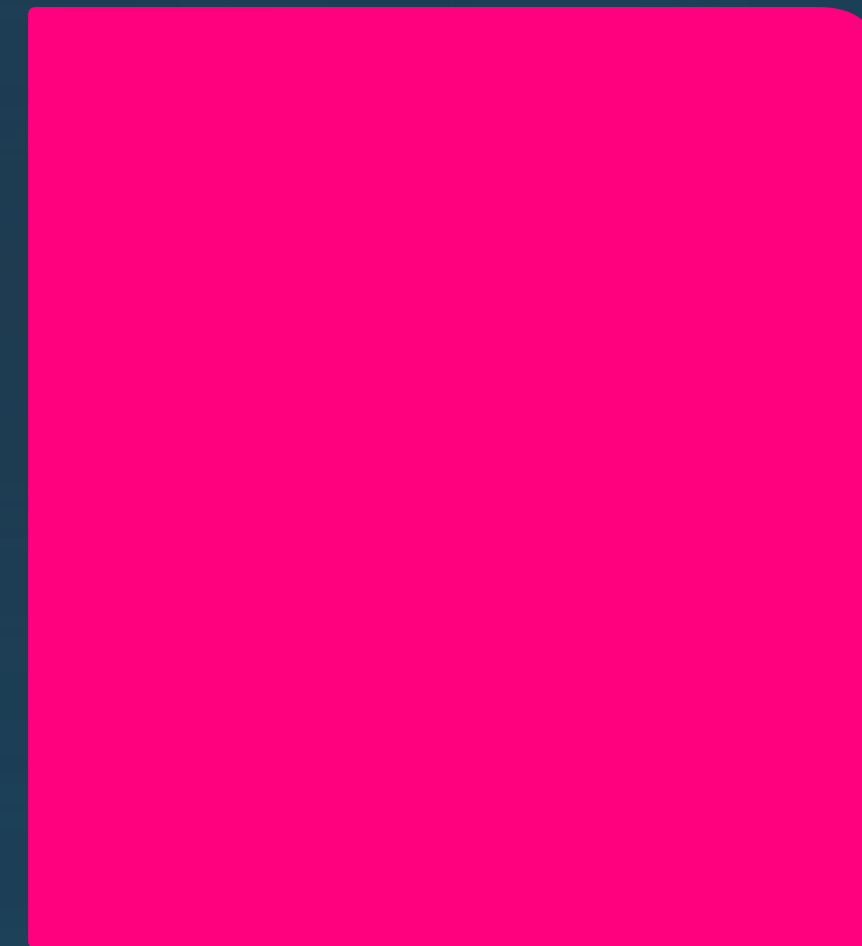
22 ✓

0 ✗

Maximize the number of elements on the interface

0 ✗

Incorporate complex patterns to challenge users



Keep the UI as simple as possible with clear names and avoid unnecessary elements

0 ✗

Use unique icons and labels to stand out

How does using **common UI** icons, labels, and patterns contribute to a positive user experience?

25 ✓



0 ✗

It adds complexity and uniqueness to the UI

1 ✗

It confuses users by presenting unfamiliar elements

0 ✗

It increases the visual appeal of the UI

What is the importance of consistency in UI design?

26 ✓

0 ✗

It limits creativity and innovation

0 ✗

It confuses users with constant changes

0 ✗

It is irrelevant in creating a successful UI

As of recommendations, how can UI designers **direct users' attention** effectively?

0 ✗

By incorporating random colors and fonts

0 ✗

By making the UI as busy as possible

2 ✗

By avoiding changes in size, placement, and color

25 ✓

By using changes in size, placement, fonts, or color of elements in the page layout

CONCLUSION

Make UI as simple as possible

- avoid unnecessary elements
- give clear names
- use common UI icons, labels, patterns
- make use of what your visitors already know to help them get their job done quickly
- be consistent
- after creating a pattern, transfer it to other parts of the app
- make the best of the page layout, color, and typography
- direct users' attention with the help of changes in the size, placement, fonts, or color of the elements
- reduce the number of unnecessary actions from a user, such as filling out the forms or searching for information
- use default values to pre-fill the forms, reduce the number of clicks to get to the most popular pages

Why is it recommended NOT to reinvent UI patterns?

2 ✕

To confuse users and challenge their adaptability

1 ✕

To make the interface unique and stand out

24 ✓

To help users learn the interface quickly and avoid frustrating experiences

What is the significance of using recognizable icons in UI design?

26 ✓

1 ✗

To increase the visual complexity of the interface

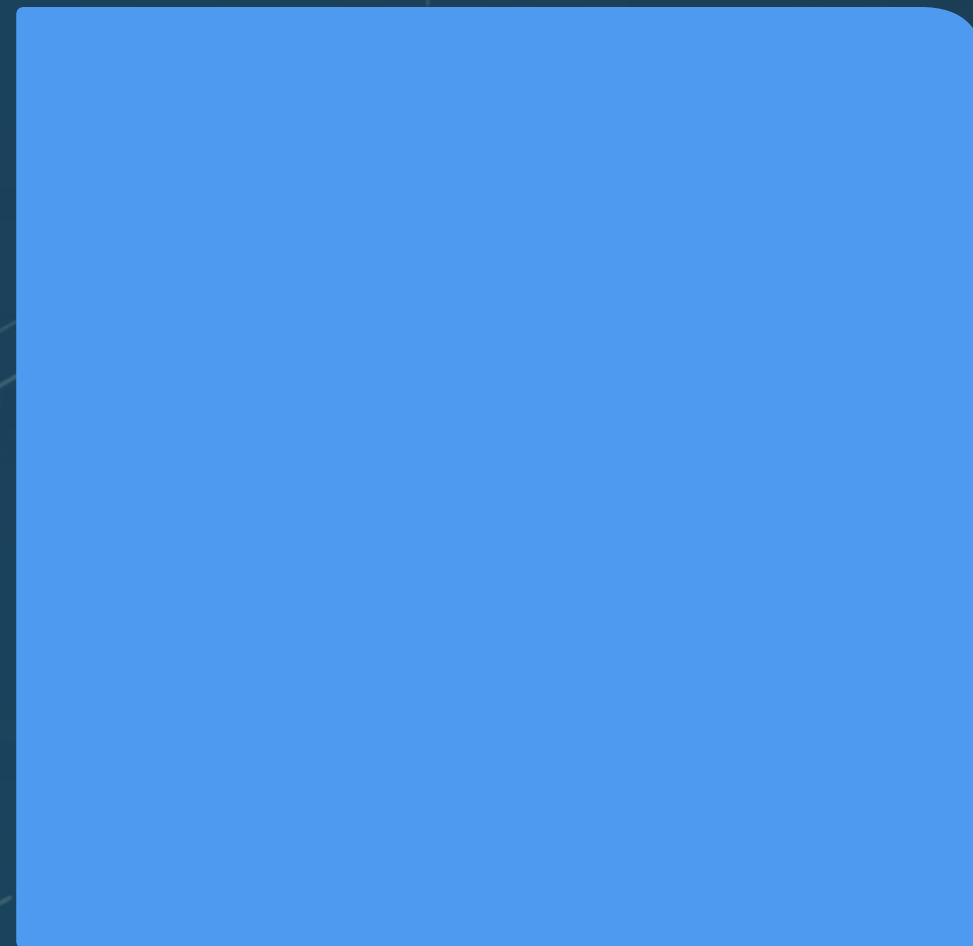
0 ✗

To challenge users with unfamiliar symbols

To leverage familiar symbols, such as an envelope for messages, for user understanding

How can designers ensure users can **easily locate** controls or information in a UI?

26 ✓



Group related objects together using color, styles (same size font), or placing them in the same box

0 ✗

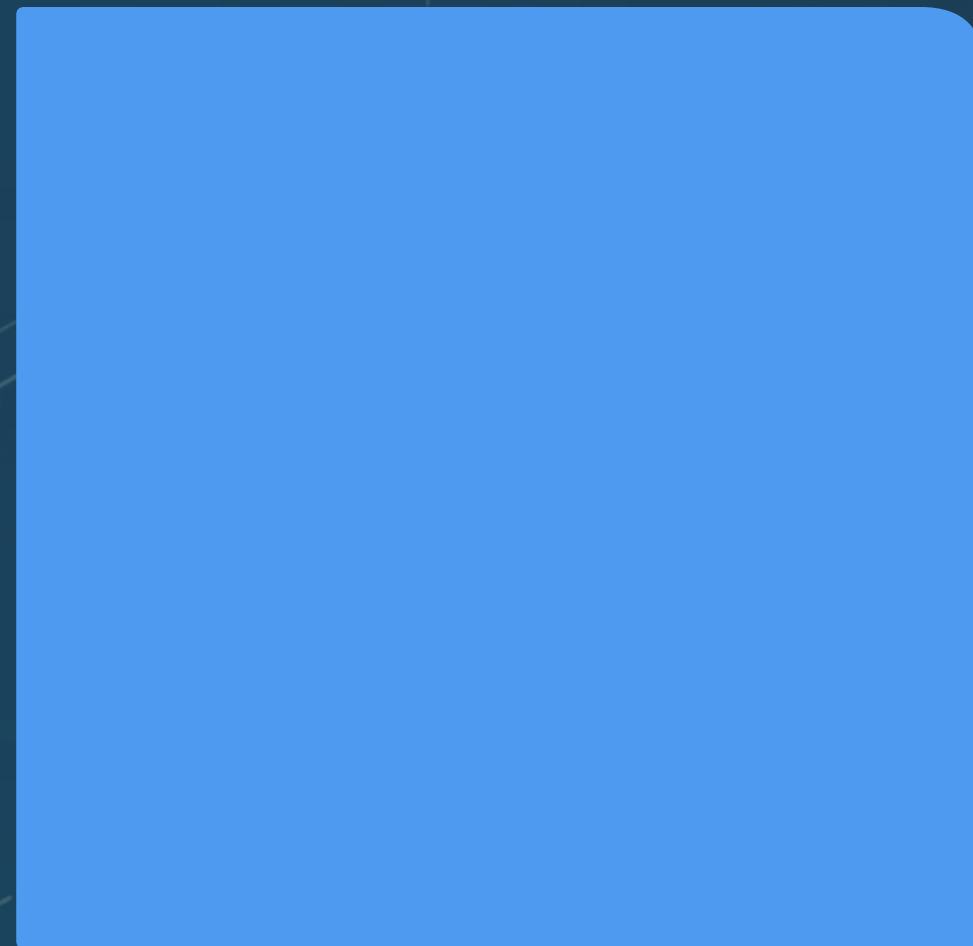
Scatter related objects across the interface to create visual interest

1 ✗

Avoid using color and styles, as they may confuse users about the grouping of elements

How can designers assist users in making decisions when encountering a new interface?

24 ✓



Add emphasis to the "next step" to guide users through their most likely task

0 ✗

Overwhelm users with multiple options to encourage exploration

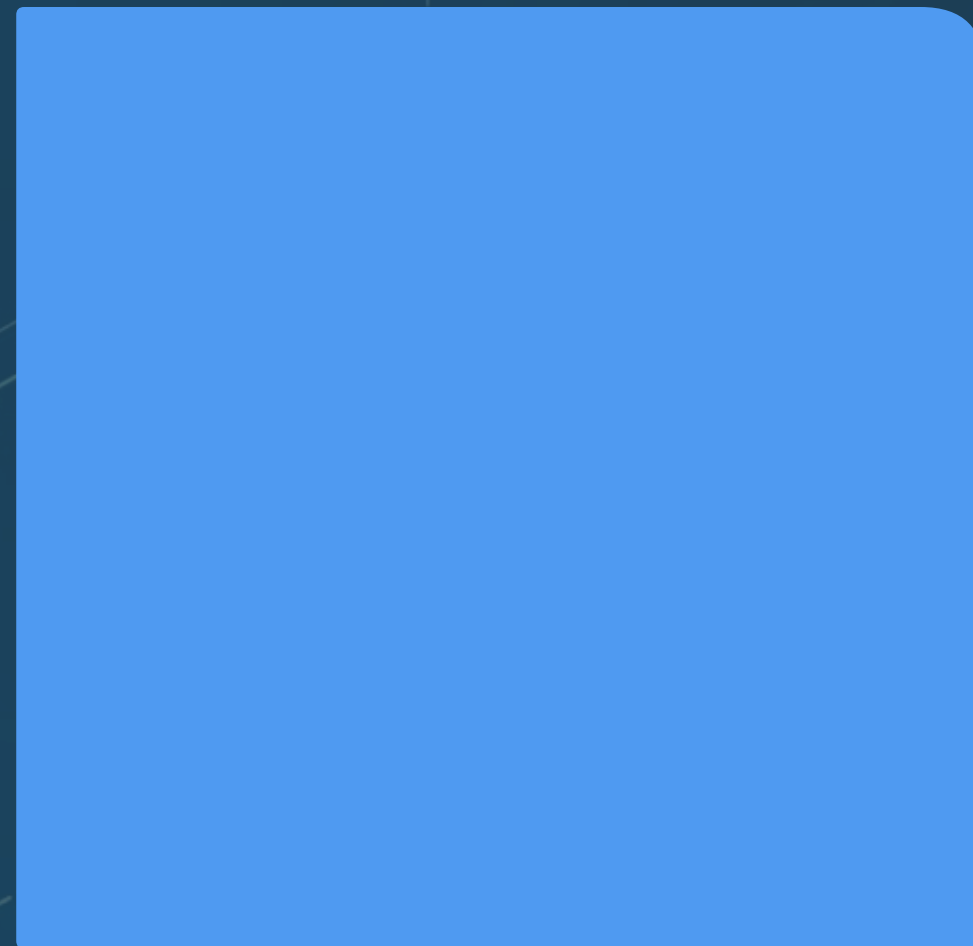
3 ✗



Include all features in the initial interface design without considering user progression

How can designers prevent users from getting lost in an app and enhance navigation?

27 ✓



Use elements like breadcrumbs or form steps to show hierarchy and eliminate unnecessary clicks

0 ✗

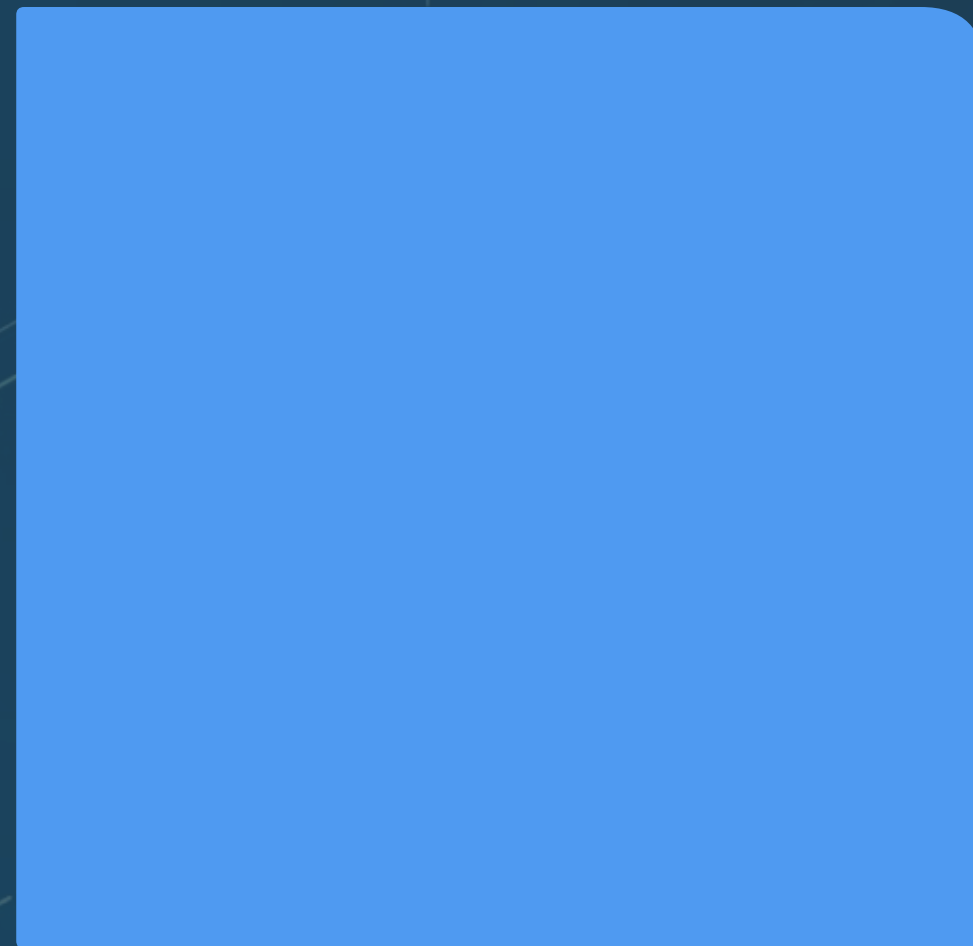
Avoid any visual cues indicating the user's location within the app

0 ✗

Rely solely on complex navigation structures for user orientation

How does optimizing loading time contribute to a positive user experience, based on the given information?

27 ✓



Users expect a web page to load in 2 seconds or less, and displaying content or a loading bar helps retain users

0 ✗

Loading time is not important for web apps

0 ✗

Consumers prefer slower loading times for a better user experience



Don't Reinvent Patterns

Group Related Elements

Less is better

Plan Before Developing

Provide Feedback

Unobtrusive Help

Help Users Decide

Focus On Target Users

Where Am I? (Use breadcrumbs)

Where did that come from?

Design For No Data

Keep consistency

Load quickly

Test

Add info buttons

<https://designforfounders.com/web-app-ux/>

Quiz leaderboard



Discuss the task

[https://elearn.university.epam.com/courses/
course-
v1:UNIVERSITY+SE124_T3+1223/coursewar
e/044040a3ecdf47bf965758eac25d3e57/5
2a998971fe740988bbf58e0e9029685/](https://elearn.university.epam.com/courses/course-v1:UNIVERSITY+SE124_T3+1223/courseware/044040a3ecdf47bf965758eac25d3e57/52a998971fe740988bbf58e0e9029685/)

Figma quick start

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026/tree/main/ws-04-architecture-advanced>

Next time: ORM. JPA.

Questions?

